

Classical–Quantum Correspondence, Schumacher Compression, and the Atypical Sector

A. Chawla

April 2, 2026

Abstract

This note develops the first step of the structured Schumacher-coding program. It establishes the classical–quantum correspondence underlying Schumacher compression, derives the typical-subspace picture from the spectral decomposition of the source state, proves the Schumacher coding theorem in a self-contained lemma-proof format, and identifies the atypical sector as a low-weight spectral tail with finite-blocklength internal structure. The purpose of this step is not only to recover the standard compression rate $S(\rho)$, but also to prepare the formal setting in which later structured refinements become natural.

1 Introduction

The first step of the program is to establish the classical–quantum dictionary that underlies structured Schumacher coding. At a conceptual level, the task is simple to state: one wishes to identify, on the quantum side, the precise analogue of the classical typical set and to recover the corresponding source-coding theorem. At a structural level, however, one also wishes to record what is discarded by the standard Schumacher procedure, because that discarded part is exactly where a later structured refinement must act.

The present step is therefore both foundational and preparatory. It develops the familiar correspondence

$$x^n \longleftrightarrow \rho^{\otimes n}, \quad A_\epsilon^{(n)} \longleftrightarrow T_\epsilon^{(n)}, \quad H(X) \longleftrightarrow S(\rho),$$

but it also keeps track of the atypical sector that lies outside the Schumacher typical subspace.

The argument proceeds in five stages:

1. The classical typical-set formalism is recalled.
2. The quantum source is described through its density operator ρ .
3. The spectral decomposition of ρ is interpreted as an induced classical source.
4. The typical subspace and Schumacher compression theorem are derived in lemma-proof form.
5. The atypical sector is analyzed as a finite-blocklength spectral tail, in a form suitable for later structured refinement.

This step establishes the source-coding layer relevant for quantum energy teleportation. It is also the point at which the later structured program receives its first formal motivation: if the atypical sector is a genuine spectral object rather than a mere error symbol, then it becomes meaningful to ask whether that sector contains additional organization that may be exploited.

2 Classical Source and Typical Set

We begin with the classical setting, since the quantum construction will ultimately mirror it through the spectrum of the density operator.

Let X be a discrete memoryless source with alphabet $\mathcal{X}$ and distribution $p(x)$. For a length- n output sequence

$$x^n = (x_1, \dots, x_n) \in \mathcal{X}^n,$$

the probability is

$$P(x^n) = \prod_{i=1}^n p(x_i).$$

The Shannon entropy is

$$H(X) = - \sum_{x \in \mathcal{X}} p(x) \log_2 p(x).$$

The ϵ -typical set is

$$A_\epsilon^{(n)} = \left\{ x^n \in \mathcal{X}^n : \left| -\frac{1}{n} \log_2 P(x^n) - H(X) \right| \leq \epsilon \right\}.$$

The role of this definition is well known. The law of large numbers implies that for large blocklength, almost all source outputs fall inside $A_\epsilon^{(n)}$, and the probability of each such output is approximately $2^{-nH(X)}$. Accordingly, the number of typical sequences is approximately $2^{nH(X)}$, and the classical source can be compressed at asymptotic rate $H(X)$ bits per symbol.

This is the key classical pattern that will reappear in the quantum problem. The quantum source will not lead directly to a classical set of words, but its spectral decomposition will produce an induced classical probability law to which the same typicality reasoning applies.

3 Quantum Source Model

Consider a memoryless quantum signal ensemble

$$\mathcal{E} = \{p(a), |\psi_a\rangle\}_{a \in \mathcal{A}},$$

where the source prepares the pure state $|\psi_a\rangle$ with probability $p(a)$ in a Hilbert space $\mathcal{H}$.

The source is represented by the density operator

$$\rho = \sum_{a \in \mathcal{A}} p(a) |\psi_a\rangle \langle \psi_a|.$$

For n independent emissions, the block source is described by

$$\rho^{\otimes n}.$$

The relevant entropy is the von Neumann entropy

$$S(\rho) = - \text{Tr}(\rho \log_2 \rho).$$

The central source-coding question is the quantum analogue of the classical one: what is the asymptotically minimal number of qubits per signal required for faithful compression of the source? Schumacher's theorem identifies this rate as $S(\rho)$ qubits per signal.

From the present perspective, however, the theorem should be viewed not only as an asymptotic coding statement but also as a structural theorem. It says that almost all of the weight of $\rho^{\otimes n}$ is concentrated on a subspace of dimension approximately $2^{nS(\rho)}$. The existence and nature of that subspace are what must now be explained.

4 Spectral Representation of the Source

The spectral representation of ρ is the mechanism through which the classical coding picture reappears inside the quantum problem. Once the source state is diagonalized, the eigenvalues of $\rho^{\otimes n}$ behave exactly like the probabilities of a classical i.i.d. source.

4.1 Illustrative Example

Consider the qubit density matrix

$$\rho = \begin{pmatrix} 0.8 & 0 \\ 0 & 0.2 \end{pmatrix}.$$

Its eigenvalues are

$$\lambda_1 = 0.8, \quad \lambda_2 = 0.2,$$

with eigenvectors

$$|e_1\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |e_2\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Hence

$$\rho = 0.8 |e_1\rangle \langle e_1| + 0.2 |e_2\rangle \langle e_2|.$$

This representation already makes visible the classical-looking aspect of the source. The eigenvectors play the role of orthogonal spectral letters, while the eigenvalues 0.8 and 0.2 supply the associated probabilities. The corresponding entropy is therefore

$$S(\rho) = -0.8 \log_2 0.8 - 0.2 \log_2 0.2.$$

The example is elementary, but its significance is general. Even when the original ensemble states $|\psi_a\rangle$ are nonorthogonal, the density operator can always be diagonalized, and its spectrum defines an induced classical distribution.

4.2 General Spectral Decomposition

In general,

$$\rho = \sum_{i=1}^d \lambda_i |e_i\rangle \langle e_i|,$$

where

- $\lambda_i \geq 0$,
- $\sum_{i=1}^d \lambda_i = 1$,
- $\{|e_i\rangle\}_{i=1}^d$ is an orthonormal eigenbasis.

Then

$$S(\rho) = - \sum_{i=1}^d \lambda_i \log_2 \lambda_i.$$

Thus the spectrum of ρ determines a classical probability distribution, even if the original ensemble states $|\psi_a\rangle$ are nonorthogonal. This is the precise sense in which the quantum source contains an induced classical source. The remainder of the coding theorem will amount to applying classical typicality to this induced law, then translating the result back into the Hilbert-space language of projectors and subspaces.

5 Typical Subspace Construction

From the spectral decomposition,

$$\rho = \sum_{i=1}^d \lambda_i |e_i\rangle \langle e_i|,$$

one obtains

$$\rho^{\otimes n} = \sum_{i_1, \dots, i_n} \lambda_{i_1} \cdots \lambda_{i_n} |e_{i_1} \cdots e_{i_n}\rangle \langle e_{i_1} \cdots e_{i_n}|,$$

where

$$|e_{i_1} \cdots e_{i_n}\rangle = |e_{i_1}\rangle \otimes \cdots \otimes |e_{i_n}\rangle.$$

The tensor-product eigenvectors therefore play the role of classical sequences, and the corresponding eigenvalues play the role of sequence probabilities. The typical-subspace construction is simply the translation of the classical typical-set formalism into this spectral language.

Definition 1 (Typical Subspace). Fix $\epsilon > 0$. The quantum ϵ -typical subspace $T_\epsilon^{(n)}$ is the span of those tensor-product eigenvectors satisfying

$$\left| -\frac{1}{n} \log_2(\lambda_{i_1} \cdots \lambda_{i_n}) - S(\rho) \right| \leq \epsilon.$$

Let $\Pi_\epsilon^{(n)}$ denote the projector onto $T_\epsilon^{(n)}$.

Equivalently,

$$T_\epsilon^{(n)} = \text{span} \left\{ |e_{i_1} \cdots e_{i_n}\rangle : 2^{-n(S(\rho)+\epsilon)} \leq \lambda_{i_1} \cdots \lambda_{i_n} \leq 2^{-n(S(\rho)-\epsilon)} \right\}.$$

This definition says that the typical subspace is generated by those spectral words whose information density remains close to the entropy $S(\rho)$. It is therefore the natural quantum replacement of the classical typical set.

Lemma 1 (Classical Source Induced by the Spectrum). *Let*

$$\rho = \sum_{i=1}^d \lambda_i |e_i\rangle \langle e_i|.$$

Then the eigenvalues of $\rho^{\otimes n}$ are exactly the products

$$\lambda_{i_1} \cdots \lambda_{i_n},$$

indexed by words $(i_1, \dots, i_n) \in \{1, \dots, d\}^n$. If I is a classical random variable with

$$\Pr(I = i) = \lambda_i,$$

then these eigenvalues coincide with the probabilities of the i.i.d. classical source

$$I^n = (I_1, \dots, I_n).$$

Proof. Using the eigenbasis of ρ ,

$$\rho^{\otimes n} = \left(\sum_{i=1}^d \lambda_i |e_i\rangle \langle e_i| \right)^{\otimes n} = \sum_{i_1, \dots, i_n} \lambda_{i_1} \cdots \lambda_{i_n} (|e_{i_1}\rangle \langle e_{i_1}|) \otimes \cdots \otimes (|e_{i_n}\rangle \langle e_{i_n}|),$$

and hence

$$\rho^{\otimes n} = \sum_{i_1, \dots, i_n} \lambda_{i_1} \cdots \lambda_{i_n} |e_{i_1} \cdots e_{i_n}\rangle \langle e_{i_1} \cdots e_{i_n}|.$$

Thus the tensor-product basis vectors are eigenvectors, with eigenvalues equal to the indicated products. On the other hand, if $I_1, \dots, I_n$ are i.i.d. with distribution $\Pr(I = i) = \lambda_i$, then

$$\Pr(I_1 = i_1, \dots, I_n = i_n) = \lambda_{i_1} \cdots \lambda_{i_n}.$$

The two lists therefore coincide. □

The lemma is the load-bearing fact behind the entire construction. It shows that quantum source coding reduces, at the spectral level, to ordinary classical typicality. Once this reduction is in place, the typical-subspace theorem follows directly from the classical asymptotic equipartition property.

5.1 Example

For

$$\rho = \begin{pmatrix} 0.8 & 0 \\ 0 & 0.2 \end{pmatrix},$$

the induced classical source has alphabet $\{1, 2\}$ with probabilities 0.8 and 0.2. For $n = 3$, the eigenvalues of $\rho^{\otimes 3}$ are

$$0.8^3, \quad 0.8^2 0.2, \quad 0.8 \cdot 0.2 \cdot 0.8, \quad 0.2 \cdot 0.8^2, \quad \dots, \quad 0.2^3,$$

which are exactly the probabilities of the corresponding classical strings. This is the spectral origin of the quantum asymptotic equipartition property.

Theorem 1 (Typical Subspace Theorem). *Fix $\epsilon > 0$. Let*

$$\rho = \sum_{i=1}^d \lambda_i |e_i\rangle \langle e_i|, \quad S(\rho) = - \sum_{i=1}^d \lambda_i \log_2 \lambda_i.$$

Then for sufficiently large n :

$$\text{Tr}(\rho^{\otimes n} \Pi_\epsilon^{(n)}) \geq 1 - \epsilon, \tag{1}$$

$$2^{n(S(\rho)-\epsilon)} \leq \dim T_\epsilon^{(n)} \leq 2^{n(S(\rho)+\epsilon)}, \tag{2}$$

$$2^{-n(S(\rho)+\epsilon)} \Pi_\epsilon^{(n)} \leq \Pi_\epsilon^{(n)} \rho^{\otimes n} \Pi_\epsilon^{(n)} \leq 2^{-n(S(\rho)-\epsilon)} \Pi_\epsilon^{(n)}. \tag{3}$$

Proof. By Lemma 1, the spectrum of $\rho^{\otimes n}$ is the probability law of an i.i.d. classical source with entropy

$$H(I) = - \sum_i \lambda_i \log_2 \lambda_i = S(\rho).$$

Applying the classical asymptotic equipartition property to this source yields (1). Each typical eigenvalue lies between $2^{-n(S(\rho)+\epsilon)}$ and $2^{-n(S(\rho)-\epsilon)}$. Since the sum of all typical eigenvalues is at most 1, the number of typical eigenvalues is at most $2^{n(S(\rho)+\epsilon)}$, proving the upper bound in (2). Since the total typical weight is at least $1 - \epsilon$ and each typical eigenvalue is at most $2^{-n(S(\rho)-\epsilon)}$, one has

$$\dim T_\epsilon^{(n)} \geq (1 - \epsilon) 2^{n(S(\rho)-\epsilon)},$$

which yields the lower bound in (2) after the usual absorption of the subexponential prefactor into the exponent. Finally, (3) is immediate because $\Pi_\epsilon^{(n)} \rho^{\otimes n} \Pi_\epsilon^{(n)}$ is diagonal in the typical eigenbasis and each diagonal entry lies in the stated interval. $\square$

The theorem expresses the full content of quantum typicality. The first inequality says that almost all of the source weight lies in the typical subspace. The second says that the dimension of this subspace grows exponentially at rate $S(\rho)$. The third makes explicit that the restricted source state is nearly flat on the typical subspace, just as the classical typical set is approximately equiprobable.

5.2 Example

For

$$\rho = \text{diag}(0.8, 0.2),$$

one has $S(\rho) = h_2(0.2) \approx 0.722$. For large n , the typical subspace is spanned by basis vectors whose index strings contain approximately $0.8n$ occurrences of the symbol 1 and approximately $0.2n$ occurrences of the symbol 2. Its dimension is therefore exponential in $nS(\rho)$ rather than in n .

6 Schumacher Compression Theorem

The typical subspace theorem yields the quantum source-coding theorem in a direct manner. Conceptually, the proof is the same as in the classical case: identify the part of the source carrying almost all the probability, count how large that part is, and embed it into a sufficiently small register.

Lemma 2 (Dimension of the Compression Register). *If a Hilbert space $\mathcal{K}$ has dimension D , then it can be embedded isometrically into a register of*

$$\lceil \log_2 D \rceil$$

qubits.

Proof. The Hilbert space of m qubits has dimension 2^m . For

$$m = \lceil \log_2 D \rceil,$$

one has $2^m \geq D$, so an isometric embedding exists. $\square$

Lemma 3 (Average Fidelity Bound). *Let the source ensemble be $\mathcal{E} = \{p(a), |\psi_a\rangle\}$, with density operator $\rho = \sum_a p(a) |\psi_a\rangle \langle \psi_a|$. Let Π be an orthogonal projector, and consider a coding map $\mathcal{N}$ that preserves the Π component perfectly and replaces the orthogonal component by a fixed fallback state. Then the average fidelity*

$$\bar{F} = \sum_a p(a) \langle \psi_a | \mathcal{N}(|\psi_a\rangle \langle \psi_a|) | \psi_a \rangle$$

satisfies

$$\bar{F} \geq \sum_a p(a) \langle \psi_a | \Pi | \psi_a \rangle^2 \geq (\text{Tr}(\rho \Pi))^2.$$

Proof. Fix a and write

$$|\psi_a\rangle = \sqrt{q_a} |t_a\rangle + \sqrt{1 - q_a} |u_a\rangle, \quad q_a = \langle \psi_a | \Pi | \psi_a \rangle,$$

where $|t_a\rangle$ is a unit vector in the range of Π and $|u_a\rangle$ is orthogonal to it. Then

$$\mathcal{N}(|\psi_a\rangle \langle \psi_a|) = q_a |t_a\rangle \langle t_a| + (1 - q_a) \tau_a,$$

for some density operator τ_a . Therefore

$$\langle \psi_a | \mathcal{N}(|\psi_a\rangle \langle \psi_a|) | \psi_a \rangle = q_a |\langle \psi_a | t_a \rangle|^2 + (1 - q_a) \langle \psi_a | \tau_a | \psi_a \rangle.$$

Since $|\langle \psi_a | t_a \rangle|^2 = q_a$ and $\langle \psi_a | \tau_a | \psi_a \rangle \geq 0$,

$$\langle \psi_a | \mathcal{N}(|\psi_a\rangle \langle \psi_a|) | \psi_a \rangle \geq q_a^2.$$

Multiplying by $p(a)$ and summing over a yields

$$\bar{F} \geq \sum_a p(a) q_a^2.$$

By convexity of $x \mapsto x^2$,

$$\sum_a p(a) q_a^2 \geq \left(\sum_a p(a) q_a \right)^2 = (\text{Tr}(\rho \Pi))^2.$$

$\square$

The lemma identifies the basic mechanism behind Schumacher coding: whenever one can project the source almost entirely into a subspace that is transmitted exactly, the average fidelity is correspondingly high. Thus the coding problem reduces to finding a sufficiently small subspace with sufficiently large source weight, and the typical subspace theorem supplies precisely that.

Proposition 1 (Direct Part). *For every $\epsilon > 0$ and sufficiently large n , there exists an encoding-decoding scheme that compresses $\rho^{\otimes n}$ into*

$$n(S(\rho) + \epsilon)$$

qubits and attains average fidelity at least $1 - \epsilon$ on the source ensemble.

Proof. Let $T_\epsilon^{(n)}$ be the typical subspace with projector $\Pi_\epsilon^{(n)}$. By Theorem 1,

$$\dim T_\epsilon^{(n)} \leq 2^{n(S(\rho) + \epsilon)}$$

for sufficiently large n . Hence there is a compression register $\mathcal{H}_C$ of size at most $n(S(\rho) + \epsilon)$ qubits and an isometry

$$V : T_\epsilon^{(n)} \rightarrow \mathcal{H}_C.$$

Define the encoder

$$\mathcal{E}(\sigma) = V \Pi_\epsilon^{(n)} \sigma \Pi_\epsilon^{(n)} V^\dagger + \text{Tr}((I - \Pi_\epsilon^{(n)})\sigma) |0\rangle \langle 0|,$$

and the decoder

$$\mathcal{D}(\tau) = V^\dagger \tau V + \text{Tr}((I - V V^\dagger)\tau) \omega,$$

where $|0\rangle$ is a fixed state in $\mathcal{H}_C$ and ω is a fixed state on $\mathcal{H}^{\otimes n}$. The composite map acts identically on the typical subspace. Therefore, by Lemma 3,

$$\overline{F} \geq \left(\text{Tr}(\rho^{\otimes n} \Pi_\epsilon^{(n)}) \right)^2.$$

Using (1),

$$\text{Tr}(\rho^{\otimes n} \Pi_\epsilon^{(n)}) \geq 1 - \epsilon$$

for sufficiently large n , hence

$$\overline{F} \geq (1 - \epsilon)^2.$$

Replacing ϵ by $\epsilon/2$ at the outset yields average fidelity at least $1 - \epsilon$. $\square$

The direct part shows that the compression rate $S(\rho)$ is achievable because the source effectively occupies only the typical subspace. Operationally, the encoder keeps the typical component intact and collapses the atypical component to a fixed fallback state. The success of the scheme is therefore entirely controlled by the weight of the atypical sector.

6.1 Example

For

$$\rho = \text{diag}(0.8, 0.2),$$

the typical subspace of $\rho^{\otimes n}$ has dimension

$$2^{nh_2(0.2)} \approx 2^{0.722n}.$$

Thus a block of n signals can be compressed into approximately $0.722n$ qubits rather than n qubits.

Proposition 2 (Converse). *For every $\epsilon > 0$ and sufficiently large n , no compression scheme into fewer than*

$$n(S(\rho) - \epsilon)$$

qubits can faithfully represent all of the typical subspace $T_\epsilon^{(n)}$.

Proof. By Theorem 1,

$$\dim T_\epsilon^{(n)} \geq 2^{n(S(\rho) - \epsilon)}$$

for sufficiently large n . A register of m qubits has dimension 2^m . If

$$m < n(S(\rho) - \epsilon),$$

then

$$2^m < \dim T_\epsilon^{(n)}.$$

Hence no injective linear map from $T_\epsilon^{(n)}$ into the compression space exists. Therefore not all orthogonal directions in the typical subspace can be preserved, and asymptotically faithful compression below rate $S(\rho)$ is impossible. $\square$

The converse complements the direct part in the usual way. The typical subspace is not merely sufficient for compression; it is also, in an asymptotic sense, the smallest subspace that can support faithful coding of the source. The rate $S(\rho)$ is therefore both achievable and necessary.

Theorem 2 (Schumacher Coding Theorem). *Let ρ be the density operator of a memoryless quantum source. Then the optimal asymptotic compression rate is*

$$R_Q = S(\rho) \quad \text{qubits per signal.}$$

More precisely:

- a) for every $\epsilon > 0$, compression at rate $S(\rho) + \epsilon$ is achievable with fidelity tending to 1 as $n \rightarrow \infty$;*
- b) for every $\epsilon > 0$, compression below rate $S(\rho) - \epsilon$ cannot achieve fidelity tending to 1.*

Proof. Part (a) is Proposition 1 and part (b) is Proposition 2. $\square$

7 The Atypical Sector and Finite-Blocklength Structure

The preceding analysis identifies the typical subspace as the component retained by Schumacher compression. For the structured program, it is equally important to identify what lies outside that subspace and how its contribution is controlled.

Definition 2 (Atypical Projector). The atypical projector is

$$Q_\epsilon^{(n)} = I - \Pi_\epsilon^{(n)}.$$

Its range

$$B_\epsilon^{(n)} := \text{Ran}(Q_\epsilon^{(n)})$$

will be called the atypical subspace.

This definition is simple, but conceptually important. The usual Schumacher construction treats the atypical subspace only through its total weight, since that is sufficient for asymptotic compression. For the structured theory, however, the atypical subspace must be regarded as an object of study in its own right.

Proposition 3 (Weight of the Discarded Sector). *For sufficiently large n ,*

$$\mathrm{Tr}\left(\rho^{\otimes n} Q_\epsilon^{(n)}\right) \leq \epsilon.$$

Proof. This is immediate from Theorem 1, since

$$\mathrm{Tr}\left(\rho^{\otimes n} Q_\epsilon^{(n)}\right) = 1 - \mathrm{Tr}\left(\rho^{\otimes n} \Pi_\epsilon^{(n)}\right) \leq \epsilon.$$

□

Corollary 1 (Quantified Information Loss Under Discard). *For the coding scheme of Proposition 1,*

$$1 - \overline{F} \leq 1 - \left(1 - \mathrm{Tr}\left(\rho^{\otimes n} Q_\epsilon^{(n)}\right)\right)^2 = 2\delta_n - \delta_n^2,$$

where

$$\delta_n := \mathrm{Tr}\left(\rho^{\otimes n} Q_\epsilon^{(n)}\right) \leq \epsilon.$$

Proof. By Proposition 1,

$$\overline{F} \geq \left(\mathrm{Tr}\left(\rho^{\otimes n} \Pi_\epsilon^{(n)}\right)\right)^2 = (1 - \delta_n)^2.$$

Rearranging yields the claim. □

The corollary makes explicit that the Schumacher procedure loses only the spectral weight carried by the atypical sector. In particular, the entire fidelity deficit is controlled by δ_n . This is precisely the quantity that later structured refinements will attempt to redistribute more intelligently across the components of the atypical subspace.

Definition 3 (Spectral Decomposition of the Atypical Sector). Let

$$\mathcal{B}_\epsilon^{(n)} = \left\{ (i_1, \dots, i_n) : \left| -\frac{1}{n} \log_2(\lambda_{i_1} \cdots \lambda_{i_n}) - S(\rho) \right| > \epsilon \right\}.$$

If

$$\mathcal{B}_\epsilon^{(n)} = \bigsqcup_{r=1}^k \mathcal{B}_{\epsilon,r}^{(n)}$$

is any partition of the atypical index set, define

$$C_r^{(n)} = \mathrm{span} \left\{ |e_{i_1} \cdots e_{i_n}\rangle : (i_1, \dots, i_n) \in \mathcal{B}_{\epsilon,r}^{(n)} \right\},$$

and let $\Pi_r^{(n)}$ be the projector onto $C_r^{(n)}$.

Proposition 4 (Orthogonal Decomposition of the Atypical Subspace). *For any partition of the atypical index set as above,*

$$Q_\epsilon^{(n)} = \sum_{r=1}^k \Pi_r^{(n)}, \quad \Pi_r^{(n)} \Pi_s^{(n)} = 0 \text{ for } r \neq s,$$

and the corresponding weights

$$w_r^{(n)} := \mathrm{Tr}\left(\rho^{\otimes n} \Pi_r^{(n)}\right)$$

satisfy

$$\sum_{r=1}^k w_r^{(n)} = \mathrm{Tr}\left(\rho^{\otimes n} Q_\epsilon^{(n)}\right) \leq \epsilon.$$

Proof. The spectral basis vectors indexed by distinct words are orthonormal. Since the sets $\mathcal{B}_{\epsilon,r}^{(n)}$ are disjoint and their union is $\mathcal{B}_{\epsilon}^{(n)}$, the associated subspaces are mutually orthogonal and their direct sum is the atypical subspace. Therefore

$$Q_{\epsilon}^{(n)} = \sum_{r=1}^k \Pi_r^{(n)}.$$

Taking the trace against $\rho^{\otimes n}$ yields

$$\sum_{r=1}^k w_r^{(n)} = \text{Tr}(\rho^{\otimes n} Q_{\epsilon}^{(n)}) \leq \epsilon$$

by Proposition 3. □

Remark 1. The proposition shows that the atypical subspace is not a featureless remainder. It is a spectral tail carrying small total weight, but it already admits canonical orthogonal decompositions indexed by subsets of atypical eigen-sequences. This observation is the minimal structural bridge from Schumacher compression to a later clustered treatment of the atypical sector.

7.1 Example: Shell Decomposition for a Diagonal Qubit Source

Let

$$\rho = \text{diag}(1-p, p), \quad 0 < p < 1.$$

For a word $(i_1, \dots, i_n) \in \{1, 2\}^n$, let m denote the number of occurrences of the symbol 2. Then every basis vector with the same value of m has the same eigenvalue

$$(1-p)^{n-m} p^m,$$

and the same multiplicity contribution

$$\binom{n}{m}.$$

Define the shell projector

$$P_m^{(n)} = \sum_{\substack{(i_1, \dots, i_n) \\ \#\{j: i_j=2\}=m}} |e_{i_1} \cdots e_{i_n}\rangle \langle e_{i_1} \cdots e_{i_n}|.$$

Then

$$\rho^{\otimes n} = \sum_{m=0}^n (1-p)^{n-m} p^m P_m^{(n)},$$

and

$$\text{Tr}(\rho^{\otimes n} P_m^{(n)}) = \binom{n}{m} (1-p)^{n-m} p^m.$$

The typical region corresponds to values of m near pn , while the atypical region corresponds to shells for which the empirical frequency m/n deviates from p beyond the prescribed ϵ -window. Hence, even before any clustering is imposed, the atypical sector already decomposes into finite-blocklength spectral shells indexed by m .

Remark 2. This shell decomposition exhibits two features relevant for the later structured program.

First, the atypical mass is not uniformly distributed across the discarded sector. Second, the dominant contributions arise from shells near the boundary of the typical window rather than from shells deep in the tail. Accordingly, finite-blocklength behavior naturally introduces a hierarchy inside the atypical sector.

8 Classical–Quantum Correspondence

The preceding analysis may be summarized in the following dictionary:

Classical source coding	Quantum source coding
sequence x^n	tensor-product eigenstate of $\rho^{\otimes n}$
probability $P(x^n)$	eigenvalue of $\rho^{\otimes n}$
typical set $A_\epsilon^{(n)}$	typical subspace $T_\epsilon^{(n)}$
atypical set $(A_\epsilon^{(n)})^c$	atypical subspace $B_\epsilon^{(n)}$
entropy $H(X)$	von Neumann entropy $S(\rho)$
compression to $nH(X)$ bits	compression to $nS(\rho)$ qubits

The principal conclusion is that classical typical-set methods and Schumacher compression are governed by the same spectral-combinatorial mechanism. At the same time, the atypical sector is identified more precisely as a low-weight spectral tail admitting orthogonal decompositions and finite-blocklength shell structure.

9 Programmatic Outlook

The role of the present step within the larger program may now be summarized succinctly.

1. **Structured Atypical Decomposition.** Step 2 will replace the single projector $Q_\epsilon^{(n)}$ by a controlled decomposition into orthogonal subspaces. The justification is provided here by the spectral resolution of the atypical sector and by the shell structure visible at finite blocklength.
2. **Choice of Quantum Geometry.** Subsequent steps will introduce metrics or divergences suitable for grouping atypical states. The justification is that the atypical sector is already represented as a family of weighted spectral components, so one may ask how these components should be organized geometrically rather than discarded uniformly.
3. **Structured Encoder and Decoder.** Later steps will refine the Schumacher encoder by attaching differentiated treatment to distinct atypical components. The justification is that Proposition 3 and Corollary 1 identify the discarded weight as an explicit error budget rather than an unstructured remainder.
4. **Dominant-Component Analysis.** The program will identify whether a small number of atypical components carry most of the residual weight. The justification is that Proposition 4 expresses the atypical mass as a sum of component weights $w_r^{(n)}$, making such a question mathematically well posed.
5. **Finite-Blocklength Refinement.** Later steps will focus on the regime in which asymptotic typicality is not yet fully dominant. The justification is that the shell decomposition for finite n already exhibits nontrivial hierarchy near the typical boundary.
6. **Limiting Recovery of Schumacher Compression.** Any structured refinement must reduce to the standard Schumacher theory when the additional organization becomes negligible. The justification is that the present step has isolated Schumacher compression as the baseline theory to which all later constructions must remain asymptotically consistent.

This outline is sufficient for the present stage. Step 1 establishes the spectral and coding-theoretic foundation, and it also indicates why the subsequent structured steps are mathematically natural.

10 Conclusion

Step 1 yields the following conclusions:

- the spectrum of $\rho^{\otimes n}$ is the probability law of an induced classical i.i.d. source,
- the typical subspace of $\rho^{\otimes n}$ has dimension of order $2^{nS(\rho)}$ and supports Schumacher compression at rate $S(\rho)$,
- the discarded component is quantified by the atypical weight $\delta_n = \text{Tr}\left(\rho^{\otimes n} Q_\epsilon^{(n)}\right)$,
- the associated fidelity loss is bounded explicitly by $2\delta_n - \delta_n^2$,
- the atypical sector already carries finite-blocklength spectral organization, providing a natural point of departure for a structured refinement.

The main contribution of this step is therefore twofold. On the one hand, it recovers Schumacher compression as the quantum counterpart of the classical noiseless coding theorem. On the other hand, it identifies the atypical sector as a mathematically structured object rather than a mere asymptotic nuisance. This is the formal setting in which the later stages of the structured program begin.